Norbert Moldován

Senior Bioinformatician

Portfolio

Highlights

10+ years in bioinformatics
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Multi-omics
-
Circulating nucleic acids
-
Python | R | Bash | PyTorch
Snakemake | Conda | Git
Slurm & HPC | Deep learning

Personal details

Date of birth: 06/02/1988
Nationality: Hu, Ro
Residence: Amsterdam, NL
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I do not require a Visa to work in the EU.

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Work experience (related)

08/2024–07/2025 | **Bioinformatician**
Erasmus University Medical Center, Department of Molecular Oncology, Rotterdam, NL
Implemented scalable cfDNA fragmentomics and multi-omics pipelines for MRD detection in cancer, and maintained HPC workflows for high-throughput sequencing data processing.

08/2020–07/2024 | **Bioinformatician**
Amsterdam University Medical Center, Cancer Center Amsterdam, Department of Pathology, Amsterdam, NL
Led cfDNA fragment end sequence analysis, developed and optimized machine learning models for liquid biopsy analysis, and built tools for circRNA detection from liquid biopsies.

09/2019–08/2021 | **Bioinformatics consultant**
Griffsoft Zrt., Szeged, HU
Advised a non-bioinformatics team on software development for a commercial metagenomic sequencing analysis tool.

Affiliations

11/2023–08/2024 | **Bioinformatician**
University of Manchester, Manchester, UK

08/2024–12/2024 | **Bioinformatician**
Amsterdam University Medical Center, Amsterdam, NL

08/2025–12/2025 | **Bioinformatician**
Erasmus University Medical Center, Rotterdam, NL

Work experience (other)

2016 | **Research assistant**
University of Szeged, Department of Medical Biology, Szeged, HU
2013–2016 | **Encounter group trainer**
Nagyító Egyesület, HU
2013–2015 | **Secondary education teacher**
Lakatos Menyhért Általános Iskola és Gimnázium, Budapest, HU
2011–2013 | **Secondary education teacher**
Jósika Miklós Elméleti Líceum, Turda, RO
2008–2013 | **Encounter group trainer**
Kolozsvári Nagyító Egyesület (Alteredu), Cluj-Napoca, RO

Education and training

2016–2020 | **PhD in Medical Genetics**
University of Szeged, Faculty of General Medicine, Szeged, HU

2010–2012 | **M.Sc. in Biology**
Babes-Bolyai University, Cluj-Napoca, RO

2007–2010 | **B.Sc. in Biology**
Babes-Bolyai University, Cluj-Napoca, RO

Mother tongue: Hungarian

Foreign languages	Understanding		Speaking		Writing
	Listening	Reading	Spoken interaction	Spoken production	
English	C2	C2	C2	C2	C2
Romanian	C1	C1	C1	C1	C1
Dutch	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user - B1 and B2: Independent user - C1 and C2: Proficient user
Common European Framework of Reference for Languages

Skills and competences

Python, R, Bash, PyTorch, TensorFlow, scikit-learn, Dask, Git, Slurm, Snakemake, Optuna, HPC, deep learning, embeddings, transformers, sequencing analysis, multi-omics, DNA, RNA, liquid biopsy, metagenomics, FAIR.

Data sources and platforms

- Proficient in analyzing short- and long-read sequencing data from Illumina, PacBio, and Oxford Nanopore technologies.
- Experienced with DNA and RNA workflows.

Machine Learning & AI in Bioinformatics

- Experienced in applying deep learning models to genomic data.
- Familiar with transformers, diffusion models, and embeddings for sequence analysis.
- Knowledgeable with scikit-learn, PyTorch and TensorFlow for bioinformatics ML applications.

High-Performance Computing & Parallelization

- Experienced in optimizing bioinformatics pipelines for multi-threading and GPU acceleration.
- Familiar with Dask for large-scale data analysis.

Scalability and reproducibility

- Proficient in Snakemake, Anaconda/Mamba for scalable and reproducible pipelines.
- Knowledgeable with ML hyper-parameter optimization with Optuna.

Communication and management

- Experienced in delivering complex concepts.
- Skilled in active listening to facilitate discussions and enhance learning outcomes.
- Confident in assertive communication, ensuring clarity and effectiveness in professional interactions.
- Proven organizational skills through planning and facilitating encounter and self-knowledge groups.
- Experienced in team leadership, guiding diverse groups as a trainer and educator.

Data management and FAIR principles

- Experienced with implementing the FAIR principles (Findable, Accessible, Interoperable, Reusable).

Grants and prizes

2024 | Dutch Research Council (NWO) | Augmenting Datasets using Diffusion Models for Liquid Biopsy Analysis.

2023 | The Netherlands Organisation for Health Research and Development (ZonMw) | Full-length Circular RNA Detection in Liquid Biopsies for Cervical Cancer Detection.

2018 | Hungarian Scientific Research Fund (OTKA) | Multiplatform analysis of herpesvirus gene expression.

2018 | Hungarian Scientific Research Fund (OTKA) | Using OMICS approach for the analysis of viral gene expression.

2018 | National Excellence Programme (UNKP) | Epstein-Barr virus and Host Cell Transcriptome Analysis

2017 | National Excellence Programme (UNKP) | AcMNPV Transcriptome Analysis with Long-Read Sequencing.

Selected publications

1. PMID: 38128532 | I developed the Fragment End Integrated Analysis (FrEIA) pipeline, combining copy number aberration, cfDNA fragment size, and end sequence diversity from liquid biopsy samples for improved cancer detection. I conducted comprehensive data analysis from raw sequencing to final results, demonstrating my ability to conceptualize innovative research bridging molecular biology, bioinformatics, and clinical oncology. This work showcases my leadership in multidisciplinary projects and expertise in developing sophisticated computational tools for complex liquid biopsy data analysis.
2. PMID: 37942753 | Led study on long (>500bp) cfDNA fragments using ONT sequencing, challenging long standing aspects of tumor-derived fragments in liquid biopsies. I directed key aspects: data curation and analysis, ITSFASTR pipeline development, visualization creation, and manuscript writing. This demonstrates my ability to drive innovative research from concept to publication in cutting-edge liquid biopsy research.
3. PMID: 39262778 | Demonstrated that EV fractions in plasma are not enriched in tumor-derived DNA, reshaping understanding of EV biology in cancer and optimizing liquid biopsy strategies. This work demonstrates my ability to address complex, long-standing questions, lead high-impact studies challenging existing paradigms, and conduct rigorous EV research.

Full publication list: Scopus | Google Scholar | ORCID

Selected packages

1. FrEIA | Comprehensive analysis of cfDNA fragment end sequences
2. ITSFASTR | All-in-one pipeline for the analysis for cfDNA fragments from ONT and Illumina sequencing
3. Ouro-tools | Pipeline for the pre-processing and analysis of circular RNA from ONT sequencing.